

① Aitken's Method

Formula:

$$\hat{p}_n = p_n - \frac{(p_{n+1} - p_n)^2}{p_{n+2} - 2p_{n+1} + p_n}$$

→ Always starts w/ p_0'

→ $\hat{p}_n$ in formula $p_0' \rightarrow$ change depend on iterations
Same, $p_0^2, p_0^3, \dots$

→ if p_n in $g(p_0')$ remain the same
change depend on iterations, p_2^0, p_3^0

Q1:

1. The following sequences are linearly convergent. Generate the first five terms of the sequence $\{\hat{p}_n\}$ using Aitken's Δ^2 method.

a. $p_0 = 0.5, \quad p_n = (2 - e^{p_{n-1}} + p_{n-1}^2)/3, \quad n \geq 1$

if asked to find 5 terms we need to find 6 p 's

⇒ apply p_n formula

$$p_0 = 0.5$$

$$\Rightarrow p_1 = \frac{(2 - e^{p_0} + (p_0)^2)}{3} = 0.2004262$$

$$p_2 = \frac{(2 - e^{p_1} + (p_1)^2)}{3} = 0.2727490$$

$$p_3 = \frac{(2 - e^{p_2} + (p_2)^2)}{3} = 0.2536071$$

$$p_4 = 0.2585503$$

$$p_5 = 0.2572656$$

$$p_6 = 0.2575989$$

⇒ apply P_6 in Aitken's formula

$$\hat{P}_n = P_n - \frac{(P_{n+1} - P_n)^2}{P_{n+2} - 2P_{n+1} + P_n}$$

⇒ Start's from $\hat{P}_0$

$$\hat{P}_0 = P_0 - \frac{(P_1 - P_0)^2}{P_2 - 2P_1 + P_0} = 0.5 - \frac{(0.2004262 - 0.5)^2}{0.2727490 - 2(0.2004262) + 0.5} = 0.2686843$$

$$\hat{P}_1 = P_1 - \frac{(P_2 - P_1)^2}{P_3 - 2P_2 + P_1} = 0.2576131$$

$$\hat{P}_2 = P_2 - \frac{(P_3 - P_2)^2}{P_4 - 2P_3 + P_2} = \underline{0.2575058}$$

$$\hat{P}_3 = P_3 - \frac{(P_4 - P_3)^2}{P_5 - 2P_4 + P_3} = \underline{0.2575911}$$

$$\hat{P}_4 = P_4 - \frac{(P_5 - P_4)^2}{P_6 - 2P_5 + P_4} = \underline{0.2575302}$$

if 4 digit or more is similar so the Solving is Correct

c. $p_0 = 0.5, \quad p_n = 3^{-p_{n-1}}, \quad n \geq 1$

⇒ apply P_n formula.

$$P_1 = 3^{-(P_0)} = 0.5773502$$

$$P_2 = 3^{-(P_1)} = 0.5303150$$

$$P_3 = 3^{-(P_2)} = 0.5584386$$

$$P_4 = 3^{-(P_3)} = 0.5414483$$

$$p_5 = 3^{-(p_4)} = 0.5516498$$

$$p_6 = 3^{-(p_5)} = 0.5455017$$

⇒ apply p_6 in Aitken's formula

$$\hat{p}_n = p_n - \frac{(p_{n+1} - p_n)^2}{p_{n+2} - 2p_{n+1} + p_n}$$

→ start from $\hat{p}_0$

$$\hat{p}_0 = p_0 - \frac{(p_1 - p_0)^2}{p_2 - 2p_1 + p_0} = 0.5481010$$

$$\hat{p}_1 = p_1 - \frac{(p_2 - p_1)^2}{p_3 - 2p_2 + p_1} = 0.5479150$$

$$\hat{p}_2 = p_2 - \frac{(p_3 - p_2)^2}{p_4 - 2p_3 + p_2} = 0.5478469$$

$$\hat{p}_3 = p_3 - \frac{(p_4 - p_3)^2}{p_5 - 2p_4 + p_3} = 0.5478225$$

$$\hat{p}_4 = p_4 - \frac{(p_5 - p_4)^2}{p_6 - 2p_5 + p_4} = 0.5478136$$

4 digit are similar thus mean is correct

② Steffen's Method

p_0^o and p_1^o mostly will be given in $\mathbb{Q}$

→ to calculate p_2^o use $g(p_1^o)$

When we use p in $g(x)$

p_1^o does not change
 $\frac{1}{t}$ will change very time

if " p_n " is given in $\mathbb{Q}$ that means in $g(x)$ form

but, if " p_n " Not given we need to Convert $f(x)$ to $g(x)$

→ apply formula

In formula p_0^1 will change
 p_0^0 not change

$$p_0^1 = p_0^0 - \frac{(p_1^0 - p_0^0)^2}{p_2^0 - 2p_1^0 + p_0^0}$$

$$p_1^1 \Rightarrow g(p_0^1) \quad \text{and} \quad p_2^1 \Rightarrow g(p_1^1)$$

$$p_0^2 = p_0^1 - \frac{(p_1^1 - p_0^1)^2}{p_2^1 - 2p_1^1 + p_0^1}$$

Q4g 4. Let $g(x) = 1 + (\sin x)^2$ and $p_0^{(0)} = 1$. Use Steffensen's method to find $p_0^{(1)}$ and $p_0^{(2)}$.

→ find p_1^0 and p_2^0 :

$$p_1^0 = g(p_0^0) = 1 + (\sin(1))^2 = 1.7080734$$

$$p_2^0 = g(p_1^0) = 1 + (\sin(p_1^0))^2 = 1.9812730$$

$$p_3^0 = g(p_2^0) = 1 + (\sin(p_2^0))^2 = 1.8407619$$

→ find p_0^1 and p_0^2 by using formula:

$$p_0^1 = p_0^0 - \frac{(p_1^0 - p_0^0)^2}{p_2^0 - 2p_1^0 + p_0^0} = \boxed{2.1529044}$$

$$p_1^1 \Rightarrow g(p_0^1) = 1 + (\sin(p_0^1))^2 = 1.6977353$$

$$p_2^1 \Rightarrow g(p_1^1) = 1 + (\sin(p_1^1))^2 = 1.9839728$$

$$p_0^2 = p_0^1 - \frac{(p_1^1 - p_0^1)^2}{p_2^1 - 2p_1^1 + p_0^1} = \boxed{1.8734640}$$

Q58

5. Steffensen's method is applied to a function $g(x)$ using $p_0^{(0)} = 1$ and $p_2^{(0)} = 3$ to obtain $p_0^{(1)} = 0.75$. What is $p_1^{(0)}$?

$$p_0^1 = p_0^0 - \frac{(p_1^0 - p_0^0)^2}{p_2^0 - 2p_1^0 + p_0^0}$$

$$\underline{0.75} = 1 - \frac{(p_1^0 - 1)^2}{3 - 2p_1^0 + 1}$$

$$\frac{(p_1^0 - 1)^2}{4 - 2p_1^0} = 1 - 0.75$$

$$\frac{(p_1^0 - 1)^2}{4 - 2p_1^0} \rightarrow \frac{0.25}{1}$$

$$(p_1^0 - 1)^2 = (4 - 2p_1^0)(0.25)$$

$$(p_1^0 - 1)^2 = 1 - 0.5p_1^0$$

$$(p_1^0)^2 - 2p_1^0 + 1 = 1 - 0.5p_1^0$$

$$(p_1^0)^2 - 2p_1^0 = \underline{-0.5p_1^0}$$

$$(p_1^0)^2 - 2p_1^0 + 0.5p_1^0 = 0$$

$$p_1^0 (p_1^0 - 2 + 0.5) = 0$$

$$p_1^0 (p_1^0 - 1.5) = 0$$

$$\boxed{p_1^0 = 0}$$

$$p_1^0 - 1.5 = 0$$

$$\boxed{p_1^0 = 1.5}$$

Q78

7. Use Steffensen's method to find, to an accuracy of 10^{-4} , the root of $x^3 - x - 1 = 0$ that lies in $[1, 2]$ and compare this to the results of Exercise 8 of Section 2.2.

$$\rightarrow f(x) = x^3 - x - 1$$

$$x^3 - x - 1 = 0$$

"from 2.2 fixed point"

$$(x^3)^{1/3} = (x + 1)^{1/3}$$

$$x = (x + 1)^{1/3}$$

$$\downarrow$$

$$g(x) = (x + 1)^{1/3}$$

$$g'(x) = \frac{1}{3} (x + 1)^{-2/3} \cdot (1)$$

$$g'(x) = \frac{1}{3} (x + 1)^{-2/3}$$

from Interval 8 $g'(a) = g'(1) = 0.2099868 < 1$

$$g'(b) = g'(2) = 0.1602499 < 1$$

$$g(x) = (x + 1)^{1/3} \Rightarrow p_n = (p_{n-1} + 1)^{1/3}$$

to find $p_0 \Rightarrow \frac{1+2}{2} = 1.5$

$$p_1^0 = g(p_0^0) = (p_0^0 + 1)^{1/3} = 1.3572088$$

$$p_2^0 = g(p_1^0) = (p_1^0 + 1)^{1/3} = 1.3308609$$

find p_0^1 and p_0^2 :

$$p_0^1 = p_0^0 - \frac{(p_1^0 - p_0^0)^2}{p_2^0 - 2p_1^0 + p_0^0} = 1.3248990$$

$$p_1^1 \Rightarrow g(p_0^1) = (p_0^1 + 1)^{1/3} = 1.3247523$$

$$p_2^1 \Rightarrow g(p_1^1) = (p_1^1 + 1)^{1/3} = 1.3247244$$

$$p_0^2 = p_0^1 - \frac{(p_1^1 - p_0^1)^2}{p_2^1 - 2p_1^1 + p_0^1} = \boxed{1.3287939}$$

$$p_1^2 \Rightarrow g(p_0^2) = 1.3254917$$

$$p_2^2 \Rightarrow g(p_1^2) = 1.3248649$$

$$p_0^3 = p_0^2 - \frac{(p_1^2 - p_0^2)^2}{p_2^2 - 2p_1^2 + p_0^2} = 1.3244180$$

$$p_1^3 \Rightarrow g(p_0^3) = 1.3246609$$

$$p_2^3 \Rightarrow g(p_1^3) = 1.3247071$$

$$p_0^4 = p_0^3 - \frac{(p_1^3 - p_0^3)^2}{p_2^3 - 2p_1^3 + p_0^3} = 1.3247179$$